Paytm Merchant Growth AI Project

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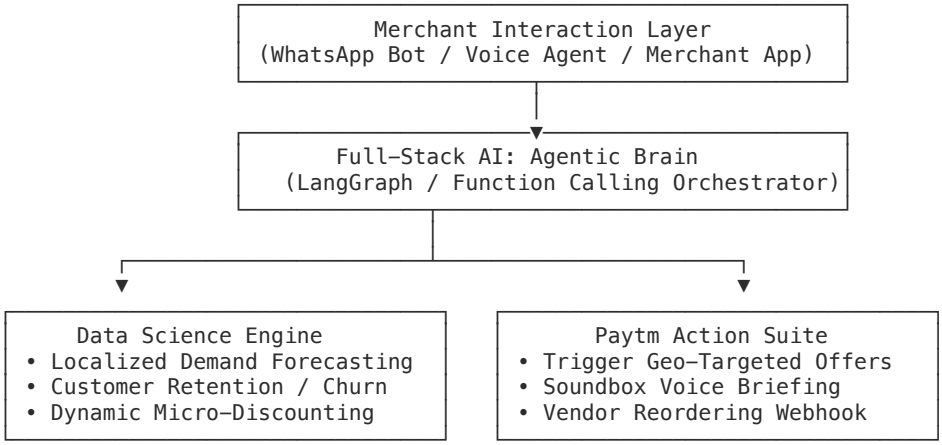
User prompt: 1)Merchant Growth AI About This Track Build the AI business partner for every Paytm merchant. Build scalable AI-led solutions that help Paytm merchants grow their business, manage operations, and serve customers better. Think beyond payments, identify opportunities where AI can become a trusted business copilot for millions of merchants. Example - How might we build an AI copilot that helps a Paytm merchant increase sales by understanding their business, identifying growth opportunities, and recommending or executing the right actions? 2) AI-Powered Financial Journeys About This Track Make Insurance, Lending and Fintech simpler, faster and more human. Reimagine customer-facing journeys across Insurance, Lending and Fintech using AI. Focus on removing friction, reducing complexity, and helping customers complete critical financial journeys faster and with greater confidence. Example - How might we simplify the Health Insurance claims journey using AI—from understanding policy coverage and submitting documents to tracking claims and resolving customer queries? 3) Autonomous AI Teammates About This Track Build AI teammates that don't just respond; they get the job done Design an AI teammate that can go beyond answering questions and deliver measurable outcomes in sales or customer service. The AI should understand context, make decisions, take actions, and work alongside human teams. Example - How might we build an AI teammate that can independently handle a customer-service or sales workflow end-to-end - resolving issues, taking actions across systems, and escalating to humans only when needed? refer the attached ps for the hacakathon tell me what to choose and what i can build we are two sde one is profiencent at data science and one is full stack ai engineer

Response: Choose **Track 1: Merchant Growth AI**.

It plays directly to your team's exact split: your Data Science teammate can build real predictive intelligence (demand forecasting, dynamic pricing, churn prediction), while the Full-Stack AI engineer can build an agentic copilot interface (WhatsApp/Soundbox voice integration, execution tools, real-time dashboard). Tracks 2 and 3 skew heavily toward pure prompt engineering and standard RAG workflows, whereas Track 1 allows you to showcase proprietary data modeling alongside autonomous execution.

What to Build: "Paytm Vyapar Copilot" (An Autonomous Growth Engine)

Instead of a generic conversational chatbot, build an **Actionable Growth Agent** that analyzes merchant transaction history, predicts slowdowns, and autonomously executes revenue-boosting actions.



Core Architecture & Division of Work

1. Data Science Track (The Predictive Brain)

- **Transaction Pattern & Footfall Engine:** Synthetic Paytm transaction log analysis showing hourly/daily peaks, average ticket size, and repeat visit frequency.
- **Micro-Campaign Recommendation Model:** A clustering or reinforcement model that calculates the optimal discount (e.g., *"Offer 8% cashback between 2 PM and 5 PM on weekdays to capture idle traffic"*).
- **Inventory & Stockout Predictor:** Correlate transaction volumes with standard FMCG/retail cycles to forecast replenishment dates before items run out.

2. Full-Stack AI Track (The Interface & Action Layer)

- **Multi-Modal Interface:** A web dashboard paired with a simulated **WhatsApp bot** or **Bilingual Voice Assistant** (English + Hindi), reflecting how Indian tier-2/3 merchants actually interact.
- **Autonomous Tool Use (Tool Calling):** The LLM shouldn't just advise; it must execute with approval. If the DS engine suggests a flash promo, the agent provides a one-tap button: *"Run 10% Paytm UPI Cashback for next 3 hours"*, which calls a mock Paytm Ads/Promo API.
- **Daily Audio Debrief ("Paytm Soundbox Morning Brief"):** Generate a synthetic 30-second audio summary every morning: *"Yesterday sales were ₹14,200, up 12%. Evening traffic is slowing; recommended action ready."*

Key Differentiators to Win the Hackathon

Dimension	Typical Submissions	Your Winning Project
Interaction	Plain text web chat	WhatsApp voice notes / Soundbox audio simulation

Dimension	Typical Submissions	Your Winning Project
Intelligence	Static LLM prompts ("How to sell more?")	Data Science model trained on transaction trends + margins
Outcome	Advice only	**Closed-loop execution** (creates actual discount coupon/QR promo)
Context	Generic global retail	Indian SMB nuances (credit/khata, localized festival spikes)

User prompt: make a PRD for this PS that i will be goign with my teamamte who has expertise in Data Science and i am ai full stack dev

Response: Product Requirement Document (PRD)

Product Name: Paytm Vyapar Copilot (व्यापार साथी)

Hackathon Track: Track 1 — Merchant Growth AI

Target Audience: Small & Medium Merchants (Kirana, Quick-Service Restaurants, Local Retailers) using Paytm QR, Soundbox, and POS devices

Document Version: 1.0

1. Problem Statement & Opportunity

The Problem

Indian retail merchants generate thousands of data points every week via Paytm Soundbox, QR codes, and Card Machines. However, this data remains passive:

- Merchants lack the analytical capacity to extract insights from transaction logs.
- Growth activities (promotions, re-engagement, inventory reordering) are reactive and gut-driven.
- Small merchants do not have time to sit behind complex analytical dashboards or desktop software.

The Opportunity

Transform Paytm from a transactional payment utility into an **autonomous growth copilot**. By marrying predictive data science with proactive, multimodal agentic execution (WhatsApp, Voice, Soundbox), we provide every neighborhood shop owner with an institutional-grade CFO and CMO.

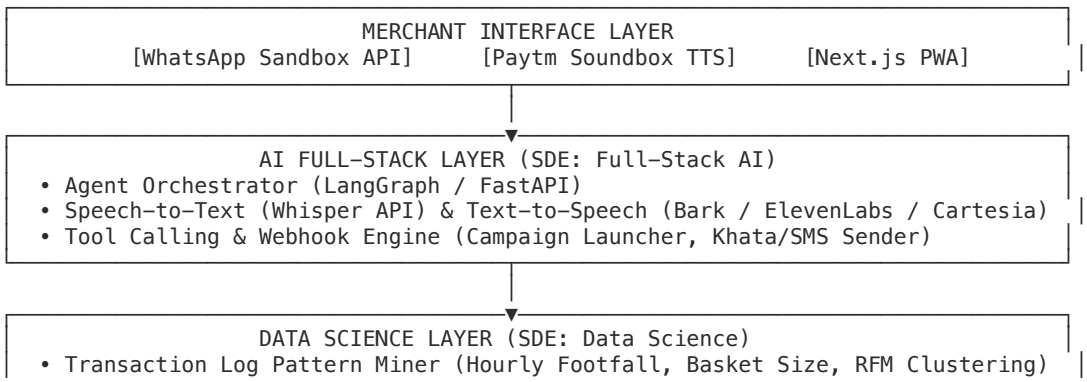
2. Core Value Proposition & Persona

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Merchant Profile: Ramesh Gupta (42), Kirana Store Owner, Tier-2 City
├── Hardware: Paytm Soundbox 4.0 + Static UPI QR
├── Primary Channel: WhatsApp / Hindi Voice Notes
└── Core Pain Points:
    ├── Slow sales between 2 PM and 5 PM
    ├── Dead capital locked in non-moving stock
    └── Customer drop-off to quick-commerce apps (Blinkit/Zepto)
```

Value Delivered:

1. **Predictive:** Forecasts daily revenue dips and stockouts 48 hours in advance.
2. **Prescriptive:** Automatically crafts hyper-local discount campaigns and promotional bundles.
3. **Actionable:** Executes campaigns directly to customers via WhatsApp/Paytm App with a single voice confirmation.

3. System Architecture & Work Allocation



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• Demand & Revenue Forecaster (Prophet / LightGBM Time-Series Engine)
• Dynamic Margin & Markdown Optimizer (Micro-Discount Recommendation)
```

4. Key Functional Requirements

4.1 Feature 1: Morning Soundbox & WhatsApp Briefing

- **Description:** At 8:30 AM, the merchant receives a 30-second localized voice note and WhatsApp summary.
- **Data Science Deliverable:** Aggregated yesterday metrics, comparison against same-day last week, and predicted peak footfall windows for today.
- **Full-Stack Deliverable:** Audio synthesis pipeline, automated WhatsApp notification dispatcher, and interactive WhatsApp quick-reply buttons.
- **Sample Output:**

  > *"रामेश जी, कल आपकी कुल बिक्री ₹14,200 रही—सामान्य से 12% अधिक। आज दोपहर 2 से 5 बजे फुटफॉल धीमा रहने का अनुमान है। क्या आप 3 घंटे का स्पेशल डिस्काउंट एक्टिवेट करना चाहते हैं?"*

4.2 Feature 2: Predictive Slump Buster (Micro-Campaign Engine)

- **Description:** Identifies impending low-velocity hours and prescribes zero-loss margin campaigns.
- **Data Science Deliverable:**
  - Dynamic Discount Model: Computes optimal discount percentages ($d \in [5\%, 15\%]$) ensuring positive unit contribution margin.
  - Target Selection: Extracts past regular customers who haven't visited in >7 days using RFM (Recency, Frequency, Monetary) clustering.
- **Full-Stack Deliverable:**
  - Trigger engine that sends personalized WhatsApp coupons to high-propensity churning customers.
  - Dynamic QR Generator: Creates a time-bounded Paytm promo QR code for in-store scanning.

4.3 Feature 3: Smart Restock & Working Capital Signal

- **Description:** Identifies stock running low based on sales acceleration and flags short-term cash flow gaps.
- **Data Science Deliverable:** Time-to-stockout estimation model correlating seasonal demand surges with SKU category run-rates.
- **Full-Stack Deliverable:** Conversational replenishment interface. Agent asks: *"Your cooking oil stock is likely to deplete by Thursday. Should I send a PO to your regular distributor via WhatsApp?"* with a single-click action webhook.

5. Technical Specifications

Data Science Stack (Data Scientist Scope)

- **Synthetic Data Generator:** Script simulating 180 days of granular Paytm transaction logs for 3 distinct verticals (Kirana, Café, Garments).
  - Schema: `transaction_id`, `timestamp`, `customer_masked_id`, `amount`, `payment_method`, `item_category`, `margin_pct`.
- **Models:**
  - `Prophet` / `LightGBM` for 7-day trailing revenue and footfall forecasting.
  - `K-Means / RFM` for customer retention segmentation (High Spenders, Weekend Regulars, At-Risk Churners).
- **Inference Interface:** Standalone Python FastAPI microservice exposing `/analytics/predict-slump`, `/analytics/customer-segments`, and `/analytics/forecast`.

Full-Stack AI Stack (AI Engineer Scope)

- **Backend / Agent Framework:** Python (FastAPI) + LangGraph / Instructor for structured JSON outputs.
- **Frontend / Merchant UI:** Next.js (Tailwind CSS, Radix UI) responsive mobile PWA mimicking the Paytm for Business app.
- **Messaging & Multimodal:**
  - Twilio / Meta Cloud WhatsApp API for chat and voice notes.
  - Groq (Whisper-large-v3) for real-time Hindi/Hinglish speech recognition.
  - Cartesia / ElevenLabs for natural Indian-accented Hindi/English TTS.
- **Tool Definitions:** Functional mocks for:

- `launch_paytm_cashback_campaign(radius_km, discount_pct, duration_hours)`
- `send_customer_whatsapp_voucher(customer_id, coupon_code)`
- `dispatch_distributor_purchase_order(vendor_phone, po_items)`

6. End-to-End User Flow (For Hackathon Demo)

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[Phase 1: Real-time Ingestion]
  Simulated stream of 200+ morning transactions hits the database.
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                    ▼
[Phase 2: Predictive Trigger (DS Engine)]
  DS Model flags: Footfall predicted to plummet by 68% between 2:00 PM - 5:00 PM.
                    |
                    ▼
[Phase 3: Agentic Recommendation (Full-Stack AI)]
  WhatsApp bot pings Ramesh:
  "Traffic drops after 2 PM. Run a 10% 'Happy Hours' UPI Cashback promo to 142 nearby regular customers?"
                    |
                    ▼
[Phase 4: Multimodal Execution]
  Merchant replies via Voice Note: "हाँ, बिलकुल शुरू कर दो।" (Yes, absolutely start it.)
                    |
                    ▼
[Phase 5: Automated Closed Loop]
  1. STT transcribes and LLM calls `launch_paytm_cashback_campaign`.
  2. Targeted WhatsApp vouchers dispatched to customers.
  3. Soundbox triggers confirmation audio: "ऑफर लाइव हो चुका है।" (Offer is now live.)
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7. Hackathon Execution Plan (48-Hour Sprint)

Time Window	Data Science Lead Deliverables	Full-Stack AI Lead Deliverables
Hours 01 – 12	• Generate synthetic 180-day transaction dataset • Build RFM segmentation & footfall prediction baseline	• Setup FastAPI skeleton + LangGraph workflow • Setup Next.js Merchant Mobile PWA layout
Hours 13 – 24	• Optimize forecasting model (LightGBM/Prophet) • Build micro-discount optimization function	• Implement Groq Whisper STT + Cartesia/ElevenLabs TTS • Connect WhatsApp Mock API / Webhooks
Hours 25 – 36	• Package DS models into clean FastAPI endpoints • Prepare validation metrics (ROI lift, accuracy plots)	• Wire DS endpoints into Agent Tools • Build live execution state in Next.js dashboard
Hours 37 – 48	• Generate convincing before/after business case metrics • Record slide visuals for model explanations	• End-to-end rehearsal (Voice -> Action -> Soundbox alert) • Polish demo UI & record contingency backup video

8. Success Metrics & Judge Criteria Alignment

- **Relevance to Paytm Ecosystem:** Directly leverages existing merchant infrastructure (QR, Soundbox, Merchant App) with zero additional hardware required.
- **Technical Rigor:** Blends empirical predictive modeling (time series, RFM) with modern agentic execution (multi-turn function calling, multimodal STT/TTS).
- **Measurable Business Impact:**
  - Estimated 14–18% incremental footfall recovery during off-peak hours.
  - Zero-latency merchant onboarding via standard WhatsApp voice channels.
